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Strain boosts propanol electrosynthesis from CO on copper

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The two-step electroreduction of CO_2 to CO followed by CO to multi-carbon products is a promising alternative to the direct CO_2 electroreduction for both efficiency and stability. The catalyst features which control selectivity in CO electroreduction remain unclear, which limits further advancement in the overall performance. Here we map the CO electroreduction reaction sensitivity to the catalyst features exploiting well-defined copper nanocrystals with tunable shape and size. After clarifying the dependence of ethylene, methane and acetate on the exposed surfaces, we reveal that copper nanospheres uniquely promote *n*-propanol selectivity. We point at the importance of strain as one of the key factors underlying the observed propanol selectivity. Inspired by these insights, we achieve *n*-propanol production via electrosynthesis with a copper-only catalyst.

The electrochemical CO reduction reaction (CORR) is an appealing catalytic reaction to produce chemicals at room temperature and ambient pressure while storing renewable electricity[1–3]. Technoeconomic analysis and experimental evidence suggest that the two-step electroreduction of CO_2 to CO followed by CO to multi-carbon products outperforms the one-step electrochemical CO_2 reduction reaction (CO_2RR) in terms of efficiency and stability because of avoiding carbonate formation[4–9]. This attractive scenario has fostered an increased effort to overcome the selectivity challenge of CORR and to improve its efficiency towards one single chemical product[10–31]. Most of the reported studies have focused on the production of ethylene and acetate[10–17]. A few studies have analysed the catalyst and environment features driving ethylene versus alcohols[1,18,19]. A notable selectivity for C_3 products has also been found in CORR, which is appealing because these efficiencies are overall consistently higher than those achieved with CO_2RR[20–31].

Interestingly, product selectivity differences towards multicarbon products in CORR and CO_2RR have emerged[8]. Differences in the microenvironment play an important role[8]. Yet, along with those, a puzzling scenario arises in relation to the catalysts features which drive selectivity in CORR, which do not appear to be the same observed in CO_2RR[1–31]. Studies on Cu single crystals have shown structural sensitivity of CORR wherein (111) and (100) surfaces preferentially evolve methane and ethylene, respectively, in CO_2RR[32,33]. Instead, two-dimensional Cu nanosheets that expose the (111) surface exhibit preferential selectivity for acetate, whereas both Cu cubes (Cu-Cubes), with mostly (100) surfaces exposed, and polycrystalline Cu nanoparticles are selective for ethylene under the same CORR conditions[14]. Surprisingly, methane has not been found as a major product in CORR. Studies show that selectivity can be tuned from acetate to ethylene by varying the CO coverage[11,12,17,34]. The formation of acetate has been strongly related to pH and catalyst macroscopic roughness[7,10]. A few studies have investigated the catalyst features driving the selectivity switch from ethylene and acetate to alcohols in CORR[19–30]. Most studies for alcohol production have focused on doping[19,25–29,31]. For example, asymmetric active sites induced by doping Cu with other metals or nitrogen have been used to explain the observed selectivity towards *n*-propanol at the expense of acetate and ethylene[25–29,31]. When introducing dopants, the deconvolution of structure and ligands effects is not trivial, hindering

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concrete conclusions on the catalyst features driving selectivity. The presence of nanoparticles with low coordinated sites with enhanced CO adsorption on metallic bulk Cu has been proposed to explain the *n*-propanol selectivity[20]. This finding contradicts some recent evidence which points at CO adsorption not being sensitive to certain structural features of the catalyst (that is, coordination numbers)[35]. Finally, interfaces between Cu(111) or Cu (110) and Cu(100) have also been suggested to favour *n*-propanol formation[21,30]. Generally, most of the catalysts used in these studies are poorly defined and are also tested in different environments and conditions, which complicates the understanding of the selectivity-driving catalyst features. Overall, solving this puzzle to connect the catalyst features to selectivity is important to elucidate reaction pathways and design catalysts which further increase selectivity in CORR.

Here, we exploit well defined Cu nanocrystals (NCs) with different shapes and sizes to map catalyst structure/selectivity relationships in CORR. We reveal that facet-dependent selectivity does exist in CORR under practical reaction conditions and impacts the production of ethylene, methane and acetate. State-of-the-art simulations identify that lattice strain boosts CORR towards alcohols and, more specifically, *n*-propanol. Consistently with strain increasing with decreasing size, we propose colloidally synthesized 4 nm Cu nanospheres, which achieves a production rate for *n*-propanol of 118 ± 4.2 mA cm^{-2}, which is ten times the current state of the art for Cu-only catalysts[20].

Results

Electrochemical CO reduction performance

Cu NCs with different shapes and size were synthesized via colloidal methods (Fig. 1a and Supplementary Fig. 1; Methods). Cu-Cube and Cu octahedra (Cu-Octa) were chosen because they expose mostly (100) and (111) surfaces, respectively, whereas Cu sphere (Cu-Sph) exhibits a mixture of both on their surface[36]. Cu-Cube with edge length of ~40 nm, Cu-Octa with edge length of ~80 nm and Cu-Sph with diameter of ~80 nm (Cu-S80) have a comparable surface-to-volume ratio (Supplementary Table 1). Cu-Sph with diameter of ~34 nm (Cu-S34) and of ~7 nm (Cu-S7) were chosen to investigate the effect of size.

Very interestingly, all tested catalysts do not show evident morphological reconstruction under the tested CORR conditions (Fig. 1a). This result contrasts with the severe restructuring occurring during CO_2RR, where the carbonates and bicarbonates in the electrolyte might play a key role in generating Cu^+ dissolving species[37–40] (Supplementary Fig. 2 and Supplementary Note 1). Although local atom mobility on the surface during catalysis cannot be ruled out, the observed morphological integrity for all shapes indicates that the main exposed surfaces are preserved.

All catalysts are selective mostly for C_{2+} products (Fig. 1b and Supplementary Table 2). The low C_1 production is in line with reported results under similar operating conditions[11–31]. The ethylene Faradaic efficiency (FE) remains constant at ~20% for all catalysts. Cu-Cube produces the highest amount of acetate (FE of ~40%). Acetate remains the main product for Cu-Octa, yet the selectivity towards this product drops (FE of ~26%) and, concomitantly, methane appears (FE of ~6%). An intriguing switch in selectivity from acetate to alcohols occurs when moving to the Cu-Sph (Fig. 1b,c). Cu-S80 shows high selectivity for C_{2+} alcohols (FE of ~50%) with *n*-propanol FE of ~20% and ethanol FE of ~29%. Alcohols increase further, whereas acetate gets reduced as the size of the spheres decreases, reaching C_{2+} alcohols (FE of ~57%) with ~23% for *n*-propanol and ~34% for ethanol for the Cu-S7. As a control, we tested commercial Cu nanoparticles, typically used in the literature[14], which possess lower alcohol production compared with Cu-Spheres of similar size (Supplementary Fig. 3).

Differences in mass transport and/or local environment among the samples can generate different distribution of multi-carbon products in CORR[7,8,10,41–43]. A similar distribution of all the catalysts through the gas diffusion layer excluded differences in mass transport among the samples as a major contributor (Supplementary Fig. 4). Furthermore, only a minor variation of the electrode potential occurs among the different systems (Supplementary Table 2 and Supplementary Fig. 5), which rules out potential dependent product distribution to explain the observed changes in selectivity[41]. Electrolyte and pH effects have been shown to impact the product distribution in CORR, with often contrasting results and more research needed on this topic[10,42,43]. However, our testing conditions are the same for all catalysts, which makes reasonable to neglect major effects of electrolyte and pH.

CO coverage can affect the CORR mechanism, and the selectivity of ethylene versus oxygenated (that is, low CO coverage to ethylene and high CO coverage to oxygenated)[11,12,17,34,44]. To explore eventual CO coverage effects, we tested our well-defined Cu NCs at different CO concentration in an inert N_2 carrier (Fig. 1d and Supplementary Fig. 6). The decrease of acetate at lower CO% in both Cu-Cube and Cu-Octa is accompanied by the increase of ethylene and methane, respectively. No change of the product distribution is observed for the Cu-S7.

The preferential ethylene selectivity at low CO coverage and the CO coverage product dependence with the switch from ethylene to acetate of the Cu-Cube from low to high CO coverage (Fig. 1d and Supplementary Fig. 6) are consistent with what observed in most of the literature[11,32,33]. The observation that methane selectivity is higher on Cu-Octa exposing the (111) surface compared with the Cu-Cube agrees with the structure-dependence observed on single crystal[32,33] (Fig. 1d). However, this selectivity is here observed for the first time at high current densities. Furthermore, the fact that this structural sensitivity becomes more evident at lower CO coverage and that methane and acetate might share a reaction intermediate on the (111) surface in CORR are insights not highlighted by previous literature.

Our experimental insight inspired us to perform density functional theory (DFT) simulations to elucidate the CORR selectivity trends on the Cu(111) surface[32,45–47] (Fig. 1e and Supplementary Note 2). A downhill pathway to methane is consistent with previous reports and with our results at low CO coverage[31,45–47]. To explain the switch from methane to acetate at higher CO coverage, we propose that the $*CH_2$ intermediate couples with *CO generating $*COCH_2$ intermediate, which leads to acetate. The lower activation energy for the OC-CH_2 coupling

Fig. 1 | Effect of catalyst size and shape on CORR. a, TEM images of the Cu catalysts as prepared (top) and after 75 min CORR (bottom): Cu-Cube, Cu-Octa, Cu-S80, Cu-S34 and Cu-S7 from left to right. Residual carbon from the gas diffusion layer (GDL) is observed in the samples after CORR as indicated in the images. **b**, Total FEs for all catalysts at 100 mA cm^{-2} in 0.5 M KOH as electrolyte. The reported data represent the mean of three independent experiments, and the error bars indicate the standard deviation (n = 3) (Supplementary Table 2). Deviation from 100% FE in the acetate selective samples can be attributed to the difficulty in detecting the acetaldehyde, being a CORR product, which transforms into acetate via a rapid non-Faradaic chemical oxidation in alkaline environments[92,93]. **c**, Partial current density for acetate, alcohols and ethylene for the tested catalysts in **b** with corresponding error bars calculated as in **b**. **d**, Normalized FE of CORR products at different CO concentration percentage in an inert N_2 carrier gas at 100 mA cm^{-2} in 0.5 M KOH for Cu-Cube, Cu-Octa and Cu-S7. The same data with error bars are reported in Supplementary Fig. 6. Mass transport effects and increased hydrogen generation dominate for all catalysts below 40% CO (Supplementary Fig. 6). The continuous lines in **c** and **d** are guides for the eyes. The loading on the gas diffusion electrodes was constant at 100 μg cm^{-2} for all catalysts. **e**, The schematic reaction pathways of CO reduction to CH_4, acetate and other C_{2+} products on Cu(111) surface with dependence on CO (%). The black straight arrows represent proton coupled electron transfer steps (ΔE), the black dashed lines represent kinetic coupling steps (E_a) and the grey arrows are used to include more than one elementary step. All energies are in electron volts and are referenced to gas phase CO_2, H_2O and H_2/H^+ energies by means of the computational hydrogen electrode[88,89] (U = −1.4 V_{SHE} and pH 14). Cu, C, O and H atoms are represented in brown, grey, red and white, respectively.

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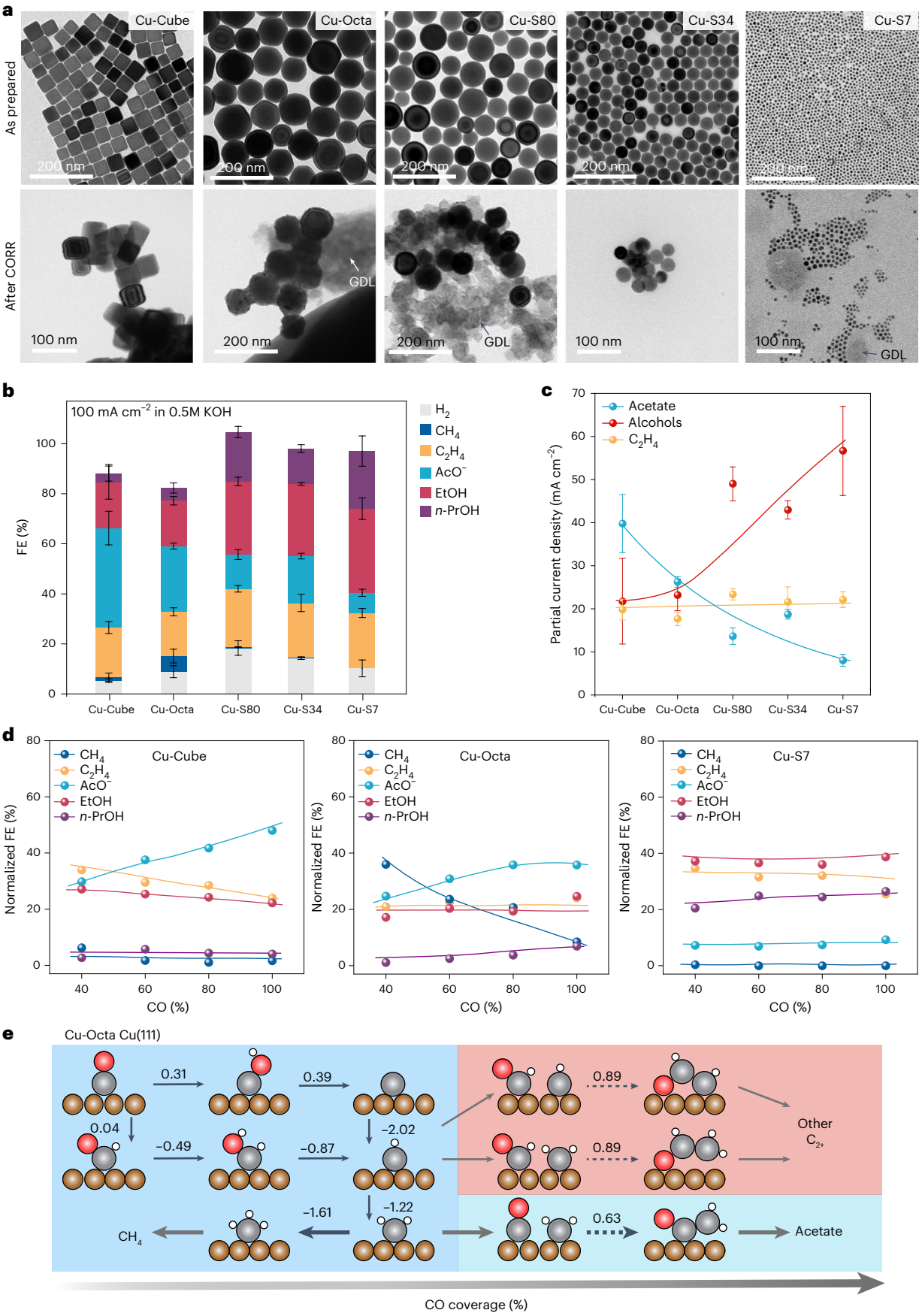
compared with other coupling reactions towards other C_{2+} products validates this path. These results point at the existence of a common $*CH_2$ intermediate between methane and acetate and that the production of one or the other depends on the CO coverage. Specifically, higher CO coverage suppresses methane production on Cu(111) surface and promotes acetate formation, which agrees with our experimental observations. Moreover, this OC-CH_2 coupling was found to be also favourable on Cu(100) (Supplementary Fig. 29a), in agreement with the



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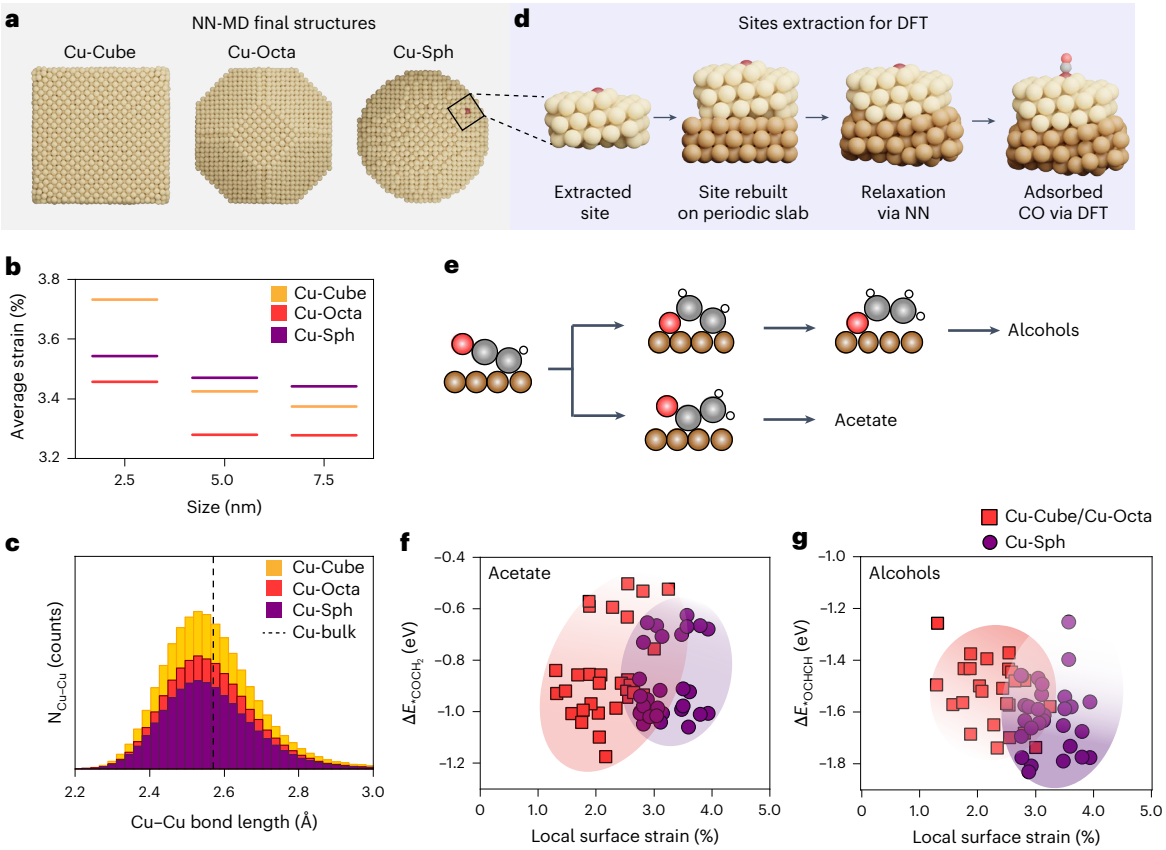

Fig. 2 | Local strain effects on CORR selectivity from theory simulations. **a**, Final structures for cube, octahedra and sphere models obtained with NN-MD. **b**, Average strain of surface Cu atoms for nine models (Supplementary equation (2)). The spheres show the highest average strain among the shapes at 5.0 nm and 7.5 nm. **c**, Histograms of Cu–Cu bond length for the three 7.5-nm models along the 334 NN-MD steps. The bulk Cu–Cu distance is represented by dashed line. The spheres exhibit the wider bond length distribution. **d**, A scheme of the site extraction and rebuilding on periodic slabs to generate structures computationally affordable in DFT without spurious electronic structure artefacts. These models were created from the last snapshot of the NN-MD simulation by extracting surface regions ($12 \times 12 \times 6$ Å^3) around the most abundant surface sites on spheres (Supplementary Figs. 17–21) and optimizing the corresponding adclusters + Cu(100) using the neural network potential (NN) (Supplementary Figs. 22–24). Cu atoms are in light brown for NCs, surface active sites are in dark red and slabs are in brown, whereas C, O and H atoms are in grey, red and white, respectively. **e**, A scheme of the studied reaction pathways leading to acetate (AcO^-), ethanol (EtOH) and propanol (*n*-PrOH), following literature[43,51]. **f**,**g**, Adsorption energies (ΔE) for $*COCH_2$ intermediate (**f**) and adsorption energies for *OCHCH intermediate (**g**) in the studied 30 active sites for cube/octahedra models (red squares) and 30 active sites for sphere models (purple dots) as a function of the local surface strain (Supplementary equation (1) and Supplementary Table 3). A more negative ΔE value indicates that the intermediate is more stable, favouring the corresponding product. The elliptical areas enclose the active sites deriving from the same models, and the colour intensity reflects the data density.

increase of acetate selectivity with CO coverage on Cu-Cube in Fig. 1d. This mechanistic insight complements the common understanding that acetate and ethylene share an intermediate in CORR[47], and further suggests that Cu(111) surfaces must be avoided for selective production of C_{2+} at diluted CO feeds. A more intriguing result emerges in Fig. 1d, which is the insensitivity of the product distribution of the Cu-S7, which possesses the highest selectivity and production rate for alcohols and, specifically, *n*-propanol.

The insensitivity of the product distribution of the Cu-S7 to the CO coverage highlights that structural effects other than the exposed surfaces must be considered to explain the unique nature of their alcohol-selective active sites. Moreover, enhanced alcohol production for the smaller Cu-Sph indicates that a size-dependent effect exists. We exclude the formation of grain boundaries as a possible explanation for the observed selectivity because of the morphological stability of the catalysts (Fig. 1a). We examined the relative differences in oxygen species among the tested catalysts during operation. Operando X-ray absorption spectroscopy (XAS) pointed at the predominance of metallic copper during CORR for all the catalysts (Supplementary Figs. 7–9 and Supplementary Note 3), in agreement with other reports[23,24,48,49]. The native ligands used for the synthesis of the catalysts also do not appear to influence the product distribution (Supplementary Fig. 10). Lattice strain can alter the binding energy of adsorbates and, thus, impact selectivity of a given catalytic reaction[50,51]. Convoluted strain and ligand effects were proposed to explain the *n*-propanol selectivity observed in Cu catalysts doped with Ag (ref. 26). Strain effects in Cu catalyst have been recently suggested as a mean to lower the activation barrier for CORR[52]. Size and shape dependence of strain in nanoparticles of noble metals more commonly used in catalysis, such as Pt, have been largely explored and different strategies implemented to modulate it, including doping/alloying and support epitaxial interactions[53–55]. Yet, how shape and size modify the strain in Cu NCs is an under investigated topic.

Theoretical simulations of catalytic surfaces

DFT cannot capture the size-dependent trend which we observe in the experiments. Thus, we used the neural network potential[56] and molecular dynamics (NN-MD) based on the ensemble of number of particles, volume and temperature (NVT) to obtain the average strain of surface atoms from models of Cu-Cube, octahedra and sphere at three different sizes (2.5 nm, 5.0 nm and 7.5 nm) (Methods; Fig. 2a,b and Supplementary Information). The average strain increases with decreasing

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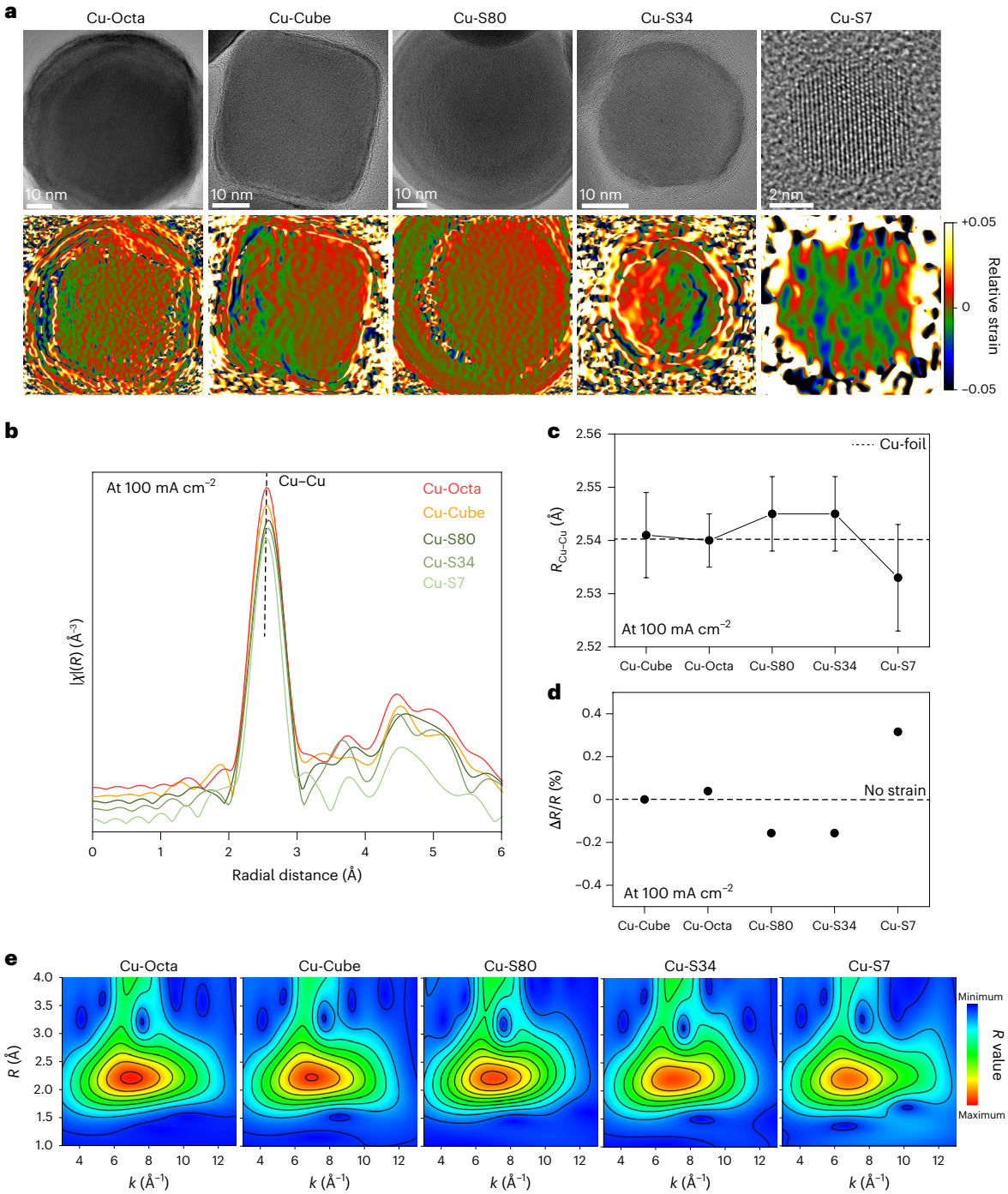

Fig. 3 | Ex situ and operando investigation of strain. a, HR-TEM images of Cu-Octa, Cu-Cube, Cu-S80, Cu-S34 and Cu-S7 with corresponding two-dimensional strain maps, which report the in-plane strain (ε_{xx}) field where the intensity scale is the relative deformation. The strain mapping is based on GPA, which extracts strain from HR-TEM images. The spots chosen for GPA are marked with yellow circles in the corresponding Fast Fourier Transform image (Supplementary Fig. 31). **b**, Operando phase-corrected EXAFS plots for Cu-Octa, Cu-Cube, Cu-S80, Cu-S34 and Cu-S7 acquired during CORR at 100 mA cm^{-2}. **c**, Cu–Cu bond length extracted from the fit of the spectra in **b**. The error bars represent the statistical uncertainties of the fitted parameters (Cu–Cu bond distance) obtained from the EXAFS least-squares fitting procedure. **d**, Estimated strain relative to the Cu-foil calculated as $100 \times \frac{R_{Cu-Cu} - R_{Cufoil}}{R_{Cu-Cu}}$. **e**, k–R WT-EXAFS contour maps where the intensity has been set to the same scale for all samples. WT-EXAFS data are not phase corrected. In **b** and **c**, the dashed line indicates the value of Cu–Cu bond length measured for the Cu foil. We note that the strain percentages obtained by GPA are closer to the values calculated by theory compared with the strain percentages obtained by EXAFS. This observation indicates that the internal referencing to the NC core is more accurate than the referencing to the Cu foil used in the EXAFS analysis. The copper foil is not an absolute standard, yet the most suitable reference under the same operating conditions used for Cu NCs. Altogether, the data indicate that Cu-S7 is the catalyst with higher strain.

size for all shapes. Such increase in strain is more pronounced for the cube and octahedra as the contribution of their highly strained edges to the overall surface area of the NCs become more significant at smaller sizes. The spheres possess the highest average strain among the shapes at 5.0 nm and 7.5 nm (Fig. 2b), which results in a wider distribution of the Cu–Cu bond length histograms (Fig. 2c).

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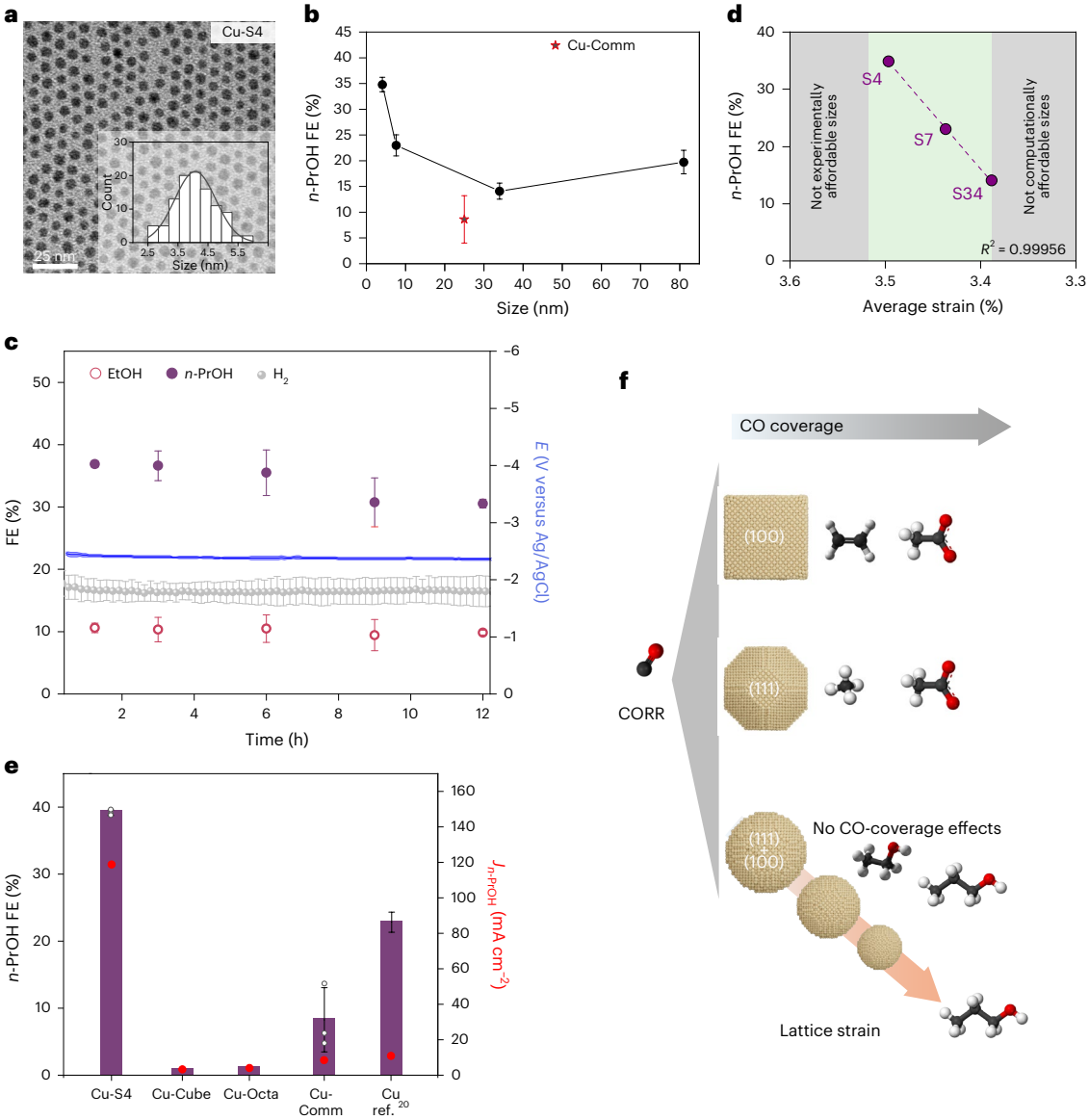

Fig. 4 | Size-dependent propanol selectivity and CORR structural sensitivity map. a, TEM image and corresponding size histogram of Cu-S4 in the inset. **b**, FE for *n*-propanol (*n*-PrOH) in Cu-Sph of different sizes tested at 100 mA cm^{-2} in 0.5 M KOH as electrolyte. **c**, FEs for *n*-PrOH, ethanol (EtOH) and hydrogen (H_2), along with the corresponding measured half-cell potential (*E*) over 12 h at a constant current of 100 mA cm^{-2} in 0.5 M KOH. The reported data in **b** and **c** represent the mean of three independent experiments, and the error bars indicate the standard deviation ($n = 3$). **d**, Experimental FE for *n*-PrOH versus the computed average strain along NN-MD obtained for spheres of 4 nm, 7 nm and 34 nm. **e**, FE and partial current density ($J_{n\text{-PrOH}}$) for *n*-PrOH of Cu-S4, Cu-Cube, Cu-Octa, Cu commercial (Cu-Comm) tested in this work at an applied current of 300 mA cm^{-2} in 1 M KOH compared with the best copper catalyst for *n*-PrOH from CORR reported so far[20]. The reported data represent the mean of three independent experiments (individual data points are reported in the graph), and the error bars indicate the standard deviation ($n = 3$). For data taken from ref. 20, the error bars correspond to those reported in ref. 20. **f**, A schematic of the general understanding of the structural sensitivity of CORR achieved using well-defined Cu nanocatalysts.

To validate our strain model and to consider the possibility of surface defect formation from dynamic local atom mobility or surface compositional changes, we intentionally and randomly removed a fraction of Cu atoms or removed and then replaced a fraction of Cu atoms with oxygen atoms (Supplementary Figs. 11–16 and Supplementary Note 4). The same trends in the strain model were observed, which corroborate the experimental conclusion that possible minor surface reconstruction and oxidation do not explain the observed CORR selectivity (see Supplementary Note 4 for more details).

To investigate how the local surface strain impacts the selectivity, we created computationally affordable active site models from the large NN-MD simulations (Fig. 2d, Supplementary Figs. 17–24 and Supplementary Note 5). This approach enables a more realistic representation of the site diversity in heterogenous catalysts compared with the conventional DFT approach of using representative slabs with one particular orientation[47,57]. Having excluded the impact of strain on CO adsorption and, thus, coverage (Supplementary Fig. 25), the as-generated active site models allowed us to assess the stability of key reaction intermediates to acetate and to alcohols (Fig. 2e). We selected these intermediates based on DFT simulations considering different C–C coupling pathways (Supplementary Fig. 26) and the identification of the key hydrogenation step determining selectivity (Supplementary Fig. 27b) and extensive literature[10,33,47,57–63]. The *CHCO intermediate, originating from CO–CO coupling or CO–COH coupling, bifurcates to *OCHCH, which then further protonates to *$OCHCH_2$, which is generally associated to alcohol production, or to *$COCH_2$, which is commonly accepted to lead to acetate[10,33,47,57–63] (Supplementary Fig. 27). We compared the adsorption energies of intermediates

of these competing pathways on 30 representative active site models extracted from the spheres and 30 extracted from cubes and octahedra (10 for each size), including a benchmark of the effect of solvation (Fig. 2f,g, Supplementary Figs. 28–30 and Supplementary Table 3). The data evidence a stronger binding towards *$COCH_2$ of the less strained surface sites of the cubes and octahedra, which is consistent with their preference for acetate. The more strained active site models of the spheres exhibit a stronger binding towards *OCHCH, which correlates the local surface strain to preference towards alcohols.

Ex situ and operando catalyst characterization

Following the insight from theory, we analysed eventual strain differences in the as-synthesized catalysts and during operation (Fig. 3 and Supplementary Note 6). Two-dimensional strain maps based on the high-resolution transmission electron microscopy (HR-TEM) images of single particles (Fig. 3a and Supplementary Fig. 31), X-ray diffraction at the ensemble level and ex situ extended X-ray absorption fine structure (EXAFS) analysis (Supplementary Figs. 32 and 33) for the as-synthesized NCs indicate a higher degree of strain for the Cu-Sph catalysts, with the Cu-S7 showing a higher overall strain (higher density of strained regions in blue colour).

Operando EXAFS analysis performed during CORR at 100 mA cm^{-2} agrees with this observation (Fig. 3b–d, Supplementary Fig. 34 and Supplementary Table 4). A shift of the Cu–Cu bond length (R) is observed in the Cu-Sph compared with Cu-Cube and Cu-Octa (Fig. 3b). The Cu–Cu bond length of Cu-Cube and Cu-Octa closely matches the one of the bulk Cu foil (2.541 Å). A deviation from this value occurs for all the Cu-Sph samples with an elongated bond length of 2.545 Å for Cu-S80 and Cu-S34 and a contracted bond length of 2.533 Å for Cu-S7 (Fig. 3c and Supplementary Table 4). This change results in a bigger relative strain ($\Delta R/R$) for the Cu-Sph compared with the Cu-Octa and Cu-Cube, with the highest value for Cu-S7 (Fig. 3d). The lowering intensity and increasing broadening of the signal in the k–R wavelet-transformed EXAFS (WT-EXAFS) contour maps going from Cu-Octa and Cu-Cube to Cu-S7 indicates a larger distribution of the Cu–Cu bond length (Fig. 3e), in agreement with the histograms from NN-MD in Fig. 2c. The overall differences among the samples are subtle because of the substantial contribution from the bulk in the XAS measurements. Therefore, these data point towards the importance of strain while not being conclusive in ruling out the impact of surface oxygen species or other surface defects in the propanol selectivity observed for the Cu-Sph.

Selective electrochemical CO reduction to *n*-propanol

Inspired by all the above insights, we hypothesized that smaller Cu NCs would exhibit an even higher selectivity for *n*-propanol. Thus, we designed the synthesis for Cu-Sph with a diameter of ~4.0 nm (Cu-S4, Fig. 4a). This catalyst exhibits a FE for *n*-propanol of 34.8% ± 1.42% (Fig. 4b, Supplementary Fig. 35a and Supplementary Table 2), which is 1.5× higher compared with that of Cu-S7 when tested under the same conditions at 100 mA cm^{-2}. Cu-S4 exhibits stable *n*-propanol production up to 12 h (Fig. 4c, Supplementary Fig. 35b and Supplementary Note 7). Operando EXAFS provides evidence of the increase in the distribution of the Cu–Cu bond length for Cu-S4 compared with Cu-S7 (Supplementary Fig. 36) along with no variation in the coordination environment and Cu–Cu bond for both samples during several hours of CORR (Supplementary Fig. 37 and Supplementary Note 7). In situ Raman performed at high current densities indicated no evident changes in the CO adsorbate binding modes, which suggests a stable surface state during operation (Supplementary Fig. 38).

Interestingly, a linear correlation emerges when plotting the experimental FE for *n*-propanol versus the computed average strain for different sizes (Fig. 4d). A comparison with commercial Cu NCs in a similar size regime indicates a unique behaviour of the colloidally synthesized Cu NCs (Fig. 4b and Supplementary Fig. 3). Unique features induced by the synthesis method itself or the wider size distribution mitigating size-dependent effects can explain the lower *n*-propanol selectivity of the commercial Cu NCs. The Cu-S4 exhibits an increase of the FE for *n*-propanol to 39.6% ± 1.4% with a corresponding partial current density of 118 ± 4.2 mA cm^{-2} when tested at 300 mA cm^{-2} (Fig. 4e and Supplementary Fig. 37a). This *n*-propanol production rate is ten times higher than the best reported Cu-only catalyst so far (Fig. 4e and Supplementary Table 5) and superior to some of those reported for Cu doped with precious and toxic metals (Supplementary Fig. 39 and Supplementary Table 5). Cu-S4 exhibits also the highest ratio between the FE of *n*-propanol and of C_2 (Supplementary Fig. 39), which corroborates further its ability to promote the coupling reaction between C_1 and C_2 towards C_3 compounds. Finally, we suggest that surface effects in pure Cu are more important than ligand effects in bimetallic catalysts for *n*-propanol formation (Supplementary Fig. 40).

Conclusion

Altogether, our results map the structural sensitivity of CORR in Cu nanocatalysts in a high current density cell via a combination of experiments and theory (Fig. 4f). The production of ethylene, methane and acetate correlates with the main crystalline surface exposed by the catalyst. Specifically, the Cu-Cube, with mostly (100) surfaces, and the Cu-Octa, with mostly (111) surfaces, selectively generate ethylene and methane, respectively, at low CO coverage. Both surfaces switch to acetate production as the CO coverage increases. On the contrary, the production of alcohols is enhanced for Cu-Sph that expose a higher number of distorted facets with low coordination sites. Local surface strain emerges as the most important feature, which drives the unique behaviour of these Cu-Sph towards propanol production.

Future operando studies might enable a more accurate quantification of surface differences among the Cu nanocatalysts and offer insight into the local mobility of surface atoms to identify the catalytic active sites along with additional direct evidence to corroborate local strain effects evidenced by theory simulations. X-ray diffraction with high energy sources might provide structural information regarding lattice strain, although assessing local atom mobility under real operating conditions at high current density remains challenging[64,65]. The findings presented here highlight the need to further develop characterization tools focused on these aspects.

Finally, the observed stability of the proposed catalysts under CORR conditions stimulates fundamental studies aimed at elucidating the underlying mechanisms and indicate the promise of these catalysts for integration into membrane electrode assembly devices properly engineered for long term operation[2,8,9,66].

Methods

Synthesis of colloidal Cu NCs

Cu NCs were synthesized following reported protocols[67,68]. Additional details on the Cu NCs synthesis and their characterization can be found in Supplementary Methods.

For Cu-Cube, tri-*n*-octylphosphine oxide (24 mmol, 9.37 g) was mixed with oleylamine (OLAM, 117 ml) in a three-neck flask and degassed under vacuum at room temperature. Copper(I) bromide (CuBr, 5 mmol, 0.71 g) was then added to the solution under N_2 flow. The resulting solution was heated to 260 °C and held at this temperature for 1 h.

For Cu-S80 and Cu-S34, tri-*n*-octylphosphine oxide was substituted by tri-*n*-octylphosphine (TOP) in the same synthetic protocol used for the Cu-Cube. For Cu-S34, TOP (0.450 ml) was added together with CuBr (86 mg) and OLAM (7 ml) in a three-neck flask, and the reaction mixture was degassed. The resulting solution was heated to 270 °C and held at this temperature for 1 h. By varying the amounts of the same reagents, Cu-S80 could be obtained. Specifically, TOP (3.5 ml), CuBr (0.92 mg) and OLAM (117 ml) were mixed in a three-neck flask, and the rest of the procedure was kept the same as described for Cu-S34.

For Cu-Octa, TOP (0.450 ml), CuBr (116 mg) and OLAM (15 ml) were added to a three-neck flask and degassed. The rest of the procedure is the same of the one reported for Cu-S34 and Cu-S80.

For Cu-S7 and Cu-S4, trioctylamine (20 ml) was added to a three-neck flask and degassed under vacuum at 130 °C for 1 h. The flask was then refilled with N_2 gas and cooled to 50 °C. Tetradecylphosphonic acid (270 mg, 1 mmol) and copper(I) acetate (245 mg, 2 mmol) were added in the flask, and the mixture was heated to 180 °C and held for 30 min. After 30 min, the mixture was heated to 270 °C and held for 30 min to obtain Cu-S7. The temperature was set to 230 °C to obtain Cu-S4, and the reaction quenched by removing the heating mantle as soon as the set temperature was reached.

Electrocatalytic measurements

Electrodes were prepared by air brushing dispersions of Cu NCs in hexane solvent on gas diffusion layer with area of 1.33 cm^2. Additional details on the electrode preparation are reported in Supplementary Information. A flow cell electrolyser was used for the electrocatalytic testing[36]. Ag/AgCl electrode (leak free series, Innovative Instruments) were used as the reference electrode, and an anionic exchange membrane (Fumasep FAB-PK-130) was interposed between the anolyte and the catholyte compartments, as counter electrode a nickel mesh (Mc Master-Carr, 100 × 100 mesh size) was used. Chronopotentiometry measurements were performed with a potentiostat Biologic SP-300. A typical experiment was run for 75 min. A gas chromatograph (GC 8610 C, SRI instruments) equipped with a HayeSep D porous polymer column, thermal conductivity detector and flame ionization detector was used for the analysis of gaseous products. Gas sampling was performed at regular intervals of 10 min. High-performance liquid chromatography or nuclear magnetic resonance was used for the analysis of the liquid products. Further experimental details can be found in Supplementary Methods.

Material characterization

Low resolution transmission electron microscopy (TEM) images were acquired on a FEI Tecnai-Spirit at 120 kV. The as-synthesized materials were drop-casted on a copper TEM grid (Ted Pella) before imaging. HR-TEM images were acquired on a double Cs-corrected FEI Titan Themis 60–300 operated at 300 kV. Geometric phase analysis (GPA) was performed in Digital Micrograph version 3.52 using a GPA plugin available at ref. 69 (ref. 70).

XAS experiments were performed at the Swiss–Norwegian beamlines BM31 at the European Synchrotron Radiation Facility in Grenoble, France. The samples were prepared in the same way of the sample used for electrocatalytic testing with a loading of 100 μg cm^{-2} and with a similar cell. Only the gas compartment was modified so that a Kapton foil could be introduced to allow X-ray penetration. The standards Cu_2O and CuO were prepared by using pressed pellets with the sample diluted in a light matrix such as boron nitride or cellulose to obtain an appropriate thickness. The measurements were carried out in fluorescence mode at an incident angle of approximately 45°. A Si(111) double crystal monochromator was used to condition the beam from the bending magnet source to a size of 3 mm (horizontal) and 300 μm (vertical).

Fluorescence X-ray absorption near edge structure spectra were acquired using a Vortex single-element silicon drift detector with XIA-Mercury digital electronics and a time resolution of 60 s per spectrum. The Cu foil standard XAS spectrum was collected in transmission using ionization chambers for transmission detection. The resulting XAS data were reduced and normalized using Athena and Larch package[71,72].

Computational details

Neural network potentials trained on 59,491 DFT points containing 1,801,491 atomic environments were employed in classical molecular dynamics (NN-MD) for Cu NCs of different shapes (cube, octahedra and sphere) and sizes (2.5, 5.0 and 7.6 nm; 34.1 only for sphere). These simulations were performed using large-scale atomic/molecular massively parallel simulator (LAMMPS) code[73] with the NN interface from a neural network potential package (n2p2)[74,75]. The NN-MDs were run for 100 ps to reach equilibrium in the canonical ensemble (NVT) using our constructed Behler–Parrinello-type HDNNP[56,76]. The sites of interest for reactivity study were selected considering coordination number and strain analysis (Supplementary Note 2) and cut from NN-MD final structure as a 12 × 12 × 6 $Å^3$-edge cube with the active site centred on the top surface. These selected active sites were further placed as adclusters on top of Cu(100) and Cu(111) three-layer surfaces using DockOnSurf[77] to avoid electronic structure embedding instabilities. DFT simulations were carried out with Vienna Ab Initio Simulation Package (VASP 5.4.4)[78,79] to compute adsorption energies of key intermediates. The exchange functional used was the Perdew–Burke–Ernzerhof[80]. Dispersion was included through the DFT-D2 method[81,82] with our reparametrized C_6 coefficients for Cu atoms[83]. Inner electrons were represented by projector augment wave[84,85], whereas the valence monoelectronic states were expanded as plane waves with a kinetic energy cutoff of 450 eV. In the final slabs, the vacuum along the *z* direction was at least 10 Å. We sampled the Brillouin zone by a Γ-centred *k*-point mesh from the Monkhorst–Pack method[86] with a reciprocal grid size smaller than 0.03 × 2π $Å^{-1}$. The slab models were asymmetric, and thus, the dipole correction was applied[87]. For all the investigated systems, structures were relaxed using convergence criteria of 0.03 eV $Å^{-1}$ and 10^{-5} eV for the ionic and electronic steps, respectively. Reported adsorption energies were obtained using CO_2(g), H_2(g), H_2O(g) and clean surfaces as energy references. The computational hydrogen electrode was used for obtaining the relative energy between H^+ and gas-phase H_2 at U = 0.0 V and pH 0 (refs. 88,89).

Data availability

Experimental data are openly available via Zenodo at https://doi.org/10.5281/zenodo.18183752 (ref. 90). Supporting DFT datasets and MD simulations are available via ioChem-BD[91] at https://doi.org/10.19061/iochem-bd-1-316. All other raw data are available from the corresponding author upon reasonable request. Source data are provided with this paper.

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Author contributions

M.W. and A.L. designed and performed experiments related to catalyst synthesis and electrocatalysis. A.L. performed all the material characterization, both ex situ and operando. E.I.-A., Z.L. and N.L. designed and carried out the theoretical part of the work. K.K. and D. Stoian provided assistance with the data collection and contributed to the analysis of electron microscopy and XAS, respectively. A.S.Z., X.L. and D. Sinton performed in situ Raman measurements. P.A., L.Z. and J.L. helped with inductively coupled plasma, NC synthesis and nuclear magnetic resonance liquid product analysis respectively and contributed with daily discussions. R.B. conceived the idea and coordinated the project. All the authors contributed to discussions and the writing of the paper.

Competing interests

There authors declare no competing interests.

Additional information

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